Questão 01

$$I = \int_0^1 \ln(1+x)\,dx$$

$$X \sim U(0,1) \quad \therefore \quad I = \int_0^1 \ln(1+x)\,dx = E[\ln(1+X)]$$

$$\hat{I}_n = \frac{1}{N}\sum_{i=1}^{N} \ln(1+x_i) \Rightarrow x_1, x_2, \ldots, x_n \sim U(0,1)$$

$$I = \int_0^1 \ln(1+x)\,dx \Rightarrow \int \ln(1+x)\,dx = (1+x)\ln(1+x) - x + C$$

$$I = \left[(1+x)\ln(1+x) - x\right]_0^1$$

$$I = \left[2\ln 2 - 1\right] - \left[1 \cdot \ln 1 - 0\right]$$

$$\ln 1 = 0$$

$$I = 2\ln 2 - 1 \quad \therefore \quad I \approx 2(0{,}69) - 1 \quad \therefore I \cong 0{,}386$$

a) $\hat{I}_n = \frac{1}{N}\sum_{i=1}^{N} \ln(1+X_i)$, $X_i \sim U(0,1)$

b) $10^2, 10^3, 10^4, 10^5$

$$\hat{I}_n = \frac{\ln(1+x_1) + \ln(1+x_2) + \ln(1+x_3) + \ln(1+x_4) + \ln(1+x_5)}{N}$$

$N = 100$

$$\hat{I}_{100} = \frac{1}{100} \sum_{i=1}^{100} \ln(1+x_i) = \frac{42,38}{100} = 0,4238$$

$N = 1000$

$$I_{1000} = \frac{1}{1000} \sum_{i=1}^{1000} \ln(1+x_i) = \frac{390,597}{1000} = 0,3906$$

$N = 10000$

$$I_{10000} = \frac{1}{10000} \sum_{i=1}^{10000} \ln(1+x_i) = \frac{3876,07}{10000} = 0,3876$$

$N = 100000$

$$I_{100000} = \frac{1}{100000} \sum_{i=1}^{100000} \ln(1+x_i) = \frac{38598,591}{100000} = 0,3859$$

$\hat{I}_n \to I$ quando $N \to \infty$

c) A simulação confirma que, à medida que o numero de amostras aumenta, a estimativa se aproxima do valor verdadeiro do integral, em concordancia com a lei de grandes numeros